

			dimensions of his printed cubes.	
2	(a)		The Principle of Conservation of Linear Momentum states that the <u>total momentum of a system remains constant</u> provided <u>no external resultant force acts on the system</u> .	B1 B1
	(b)		An elastic collision between two or more objects is one in which <u>kinetic energy is conserved</u> .	B1
	(c)		initial momentum = $(0.400)(5.0) = 2.0 \text{ kg m s}^{-1}$ momentum of P after collision = $(0.400)(-0.40) = -0.16 \text{ kg m s}^{-1}$ momentum of Q after collision = $2.0 - (-0.16) = 2.16 \text{ kg m s}^{-1}$	M1 A1
	(d)		For elastic collision, relative speed of approach = relative speed of separation $u_P - u_Q = v_Q - v_P$ $5.0 - 0 = v_Q - (-0.40)$ $v_Q = 4.6 \text{ m s}^{-1}$ $m_Q v_Q = 2.16 \text{ kg m s}^{-1}$ $m_Q = (2.16) / (4.6) = 0.47 \text{ kg}$	M1 A1
	(e)		$F = \Delta p / \Delta t$ $= (2.16 - 0) / (0.060)$ $= 36 \text{ N}$	A1
3	(a)	(i)	The positive direction was defined to be the direction from A to	